Digital Transformation Accelerator

Inception Report

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Table of Contents

Acronyms

AI	Artificial Intelligence
AoW	Area of Work
ARTC	Abu Dhabi Technology and Research Center
CGIAR	Consultative Group on International Agricultural Research
CWANA	Central, Western Asia & North Africa
DTA	Digital Transformation Accelerator
DI	Digital Innovation
FAIR	Findable, Accessible, Interoperable, and Reusable
FLW systems	Food, Land, and Water Systems
ISDC	Independent Science for Development Council
HLO	Higher Level Outputs
I-OC	Immediate Outcome
KASH	Knowledge, attitudes, skills, and habits
LAC	Latin America and the Caribbean
MELIA	Monitoring, Evaluation, Learning, and Impact Assessment
ToC	Theory of Change
PCU	Portfolio Coordination Unit
PRMF	Performance and Results Management Framework
WCA	West and Central Africa

Executive summary

The Digital Transformation Accelerator (DTA) is one of 13 science programs and accelerators in the CGIAR's 2025–2030 Science and Innovation Portfolio. DTA is designed to harness digital and AI technologies to address challenges in food, land, and water (FLW) systems, with a focus on equitable and ethical innovation. DTA's pathway to achieve these outcomes is by accelerating CGIAR science and innovations by co-designing, co-creating and embedding inclusive and ethical digital innovations into innovation workflows of other science programs, accelerators and the broader FLW systems. Through an iterative and consultative process with the complete spectrum of actors in the FLW innovation systems, DTA identified four areas of work (AoWs) that can deliver the most impactful outcomes. The four interconnected AoWs are: Data Ecosystem, Action Lab, Digital Futures, and Enabling Environment. These AoWs collectively aim to enable Findable, Accessible, Interoperable, and Reusable (FAIR) and AI-ready data ecosystems, co-develop inclusive digital innovations, anticipate emerging technological shifts, and create enabling policy and governance frameworks.

The Digital Innovation Initiative (DI) which was part of the P22 portfolio from 2022 to 2024 created a strong foundation for DTA and achieved significant milestones, including the development of over 60 digital innovations and 236 knowledge products across 47 countries, training over 2,500 individuals, and supporting inclusion-focused digital access through partnerships with over 250 organizations. DTA will continue to build on the momentum, success and partnerships of DI and accelerate CGIAR's transformation towards an AI and digitally ready organization. Since December 2024, the DTA transition team has worked through the inception phase and has engaged with other science programs and accelerators as well as key partners and stakeholders to rethink and revise DTA's long- and short-term goals, work plans, objectives, research questions, Theory of Change (ToC) as well as MELIA plan. This document provides a concise summary of these key inception workflows, processes, insights and outputs, as developed by the transition team during the inception phase.

During the Inception Phase, DTA refined its ToC based on the Independent Science for Development Council (ISDC) feedback and extensive stakeholder engagement, including co-design workshops with science programs, accelerators, and tech partners like Microsoft, Google, and Amazon Web Services (AWS). The Accelerator identified nine high-level outputs (HLOs) at the accelerator level. A tailored MELIA framework was developed to track progress across the 23 HLOs and 12 intermediate outcomes, ensuring alignment with CGIAR's system-level goals. A rigorous prioritization process identified regional focus areas, including Latin America and the Caribbean (LAC) for biodiversity and inclusion, Central, Western Asia & North Africa (CWANA) for precision irrigation, and South Asia for nutrition security. The final Accelerator design integrates quantitative rankings, partner priorities, donor mandates, and operational feasibility, resulting in strategic deviations to enhance impact and responsiveness. DTA's comparative advantage analysis informed a federated partnership model where CGIAR's domain expertise complements the technical prowess of external partners. Approximately 60 percent of planned infrastructure investments were reallocated to support capacity building and co-development. Cross-portfolio linkages were investigated further and joint use-cases opportunities identified. These use cases were also a critical input for the revised HLOs, ToC and MELIA plan. DTA continues to acknowledge the global nature of its work but is intentional in creating local capacities and innovation ecosystems through network of hubs. Key emerging risks for data and digital innovations and mitigation measures were identified during the inception phase. As DTA transitions to the implementation phase, it remains committed to fostering inclusive, responsible digital transformation through co-created innovations, robust partnerships, and strategic foresight - positioning CGIAR and its partners to accelerate scientific impact in service of global food system resilience and equity.

1. Program/ Accelerator overview

1.1 High-level vision

DTA's vision remains consistent and strongly aligned as laid out in the original proposal document. Its overarching vision is to enable CGIAR, and partners to harness the AI and digital revolution to accelerate the design, development and deployment of science and innovations to address complex and interconnected challenges of climate change, malnutrition and income inequalities. To this end, DTA will co-develop digital tools and innovations that accelerate scientific discovery in FLWs, facilitate evidence-based decisions, as well as create the necessary human and institutional capacities. However, digital innovations (especially AI) when applied without due considerations to the issues of equity, ethics, human centered design and principles of responsible innovation could lead to perverse outcomes including the widening of digital divide and asymmetric power distributions. DTA will therefore also prioritize creating global public goods including FAIR and AI-ready datasets that will support digital innovations for equitable and sustainable FLW transitions. DTA also envisions developing critical capacities within CGIAR and partners to foresight AI and digitally enabled transitions and guide future investments and strategies for equitable and inclusive outcomes. Some critical science/ research questions that DTA will address during the implementation include incentives for mainstreaming FAIR and AI-ready data ecosystems; inclusive, ethical and human centric digital innovations; emerging technologies shifts and institutional frameworks, policies, and innovation networks

1.2 Achievements from 2022 to 2024

Between 2022 and 2024, the CGIAR Initiative on Digital Innovation addressed three major barriers to digital transformation in agrifood systems: weak information systems, the digital divide, and low levels of digital literacy and skills.

To address weak information systems, the Initiative demonstrated how digital innovations can generate timely, actionable data. In southern Africa, it co-created a "Digital Twin" of a regional ecosystem. In Odisha, India, advisory services were expanded to 6,000 farmers, while in Kenya, livestock market intelligence tools reached 4,000 pastoralists. In Rwanda and Guatemala, a crowdsourced approach was used to generate real-time diet quality data at low cost, and in Mozambique, satellite imagery provided an alternative source of agricultural production statistics in collaboration with the government.

To bridge the digital divide, especially for women and rural communities, the Initiative supported five organizations through the Inspire Challenge to apply human-centered design and increase women's use of digital agriculture services. It developed a new tool to help service providers assess and improve the inclusivity of their platforms. In parallel, AI tools were used to support Farm Radio International in translating, transcribing, and analyzing farmer calls in Swahili and Luganda, enhancing the reach of audio-based advisory systems.

In response to low digital literacy and skill levels, the Initiative developed digital tools, participatory platforms, and learning partnerships. Two editions of the ICTforAg conference and the launch of the ICTforAg Learning Network engaged over 800 stakeholders. Fully digital plots and ranches were established in Guatemala and Kenya to build local capacities. AI applications were explored to make technology more accessible. For example, the Water CoPilot in the Limpopo River Basin and a citizen science app with machine learning, enabled record verification.

Across these areas, the Initiative generated over 20 use cases, 60 new innovations, 236 knowledge products, and 95 additional outputs, contributing to 28 cases of innovation adoption, 16 policy changes, and 14 other reported outcomes. It trained 2,533 individuals (31 percent of them women) and conducted 45 capacity-sharing initiatives, ranging from professional diploma courses to WhatsApp-based trainings for farmers.

Operating in 47 countries and partnering with over 250 organizations, the Initiative demonstrated the power of strategic collaboration across public, private, research, and international sectors. Government partnerships helped create enabling environments, private-sector actors linked research with markets, research institutions ensured scientific rigor, and NGOs supported inclusion, particularly in gender and digital literacy. This multi-sectoral approach enabled the development, uptake, and scaling of responsible digital innovations, laying the foundation for a more inclusive and sustainable digital transformation of agrifood systems.

1.3 Description of intended research

Area of Work (AoW) 1: Data Ecosystem

Data Ecosystem will strengthen the foundations for data-driven research and innovation across CGIAR by addressing two core research questions:

1. How CGIAR and its partners can effectively leverage data and analytics to accelerate research and digital services.
2. How knowledge, attitudes, skills, and habits (KASH) can be optimized to deliver AI-ready data and solutions.

The overarching aim is to make agricultural research data open, FAIR, and AI-ready; to enable users to access interoperable datasets and tools; to mobilize data through robust analytics pipelines; and to promote a culture of responsible and inclusive data use.

To achieve these goals, Data Ecosystem will undertake a range of research and support activities. These include the development of tools to monitor FAIR compliance, the deployment of standardized data collection systems, the establishment of scalable analytics platforms and computing infrastructure, and the implementation of data governance frameworks. Complementary efforts will focus on capacity development, technical support, and culture change, along with stakeholder engagement through human-centered design approaches that ensure inclusivity and relevance.

Data Ecosystem will be organized around four interdependent flagships. The first flagship, the Data Collaborative will support shared data governance, the co-creation of standards, and a cross-institutional culture of ethical, inclusive data stewardship. A second flagship will establish a Reference Architecture - a federated infrastructure that enables the integration of data, processes, and services across CGIAR. A third flagship will focus on scaling agricultural management systems (AMS) to support the collection of harmonized, FAIR-compliant agronomic data. The fourth will address the harmonization of historical datasets, helping CGIAR centers identify, clean, and convert legacy data into formats that are FAIR and ready for advanced analytical use.

The expected outputs of Data Ecosystem include improved metadata standards, interoperable data tools, accessible digital platforms, and targeted training resources. These are anticipated to lead to intermediate outcomes such as increased generation and use of FAIR, AI-ready data; stronger institutional capacity for responsible data use; and the cultivation of a data culture across CGIAR. Ultimately, Data Ecosystem will contribute to CGIAR's long-term outcomes by supporting equitable access to digital tools, enabling cross-sectoral collaboration, and advancing the institutionalization of FAIR principles within FLW systems.

AoW 2: Action Lab
Action Lab will work towards accelerating the adoption of digital solutions and equitable access to AI-enabled tools in FLW research through cutting-edge, interdisciplinary research at the intersection of AI, agrifood systems science, and inclusive innovation. Action Lab will prioritize innovative, use case-driven research that goes beyond traditional research approaches to test, adapt, and scale AI-powered tools within agrifood systems. Action Lab's priority use cases will be co-designed with CGIAR Science Programs and Accelerators, ensuring alignment with the CGIAR Research Portfolio's Theory of Change. To further amplify its impact, Action Lab will establish high-profile technical partnerships with global leaders in AI and digital technologies. These partnerships will complement CGIAR's deep domain expertise with partners' cutting-edge capabilities in areas such as machine learning, large language models, digital infrastructure, and responsible AI practices to create synergies that accelerate innovation, scaling, and impact. This research will involve co-developing and piloting AI-powered analytics and decision-support tools across diverse domains. It will require methods such as machine learning model training and fine-tuning, causal inference and time-series forecasting, and natural language processing for research, advisory, and learning platforms. AI models will be embedded into research processes, iteratively tested with users, and refined using feedback loops. To ensure inclusive outcomes, the research will apply human-centered design and participatory methods, such as co-creation workshops with extension agents and farmers, A/B testing of digital advisory formats, and disaggregated adoption studies to understand access barriers for women, youth, and marginalized producers. The research will generate the evidence, tools, and frameworks needed to guide responsible digital transformation in agrifood systems.

AoW 3: Digital Futures
Digital Futures focuses on identifying and exploring emerging digital technologies and future scenarios with the potential to transform FLW systems. Its research aims to help CGIAR and partners anticipate technological shifts, critically assess their implications, and strategically identify opportunities for CGIAR. The anticipated outputs from this AOW could be particularly relevant in the context of LLMs and Generative AI which are rapidly evolving and enable CGIAR to adapt and navigate this complex domain.

The intended research includes:

- Horizon scanning and foresight to monitor and synthesize trends in AI, and other frontier technologies relevant to CGIAR's mission.
- Technology probes: small, rapid experiments to test early-stage digital innovation
- Futures scenario development, co-designed with experts on possible socio-technical directions and their implications
- Strategic intelligence and investment guidance for digital innovation, supporting prioritization and responsible scaling of digital solutions.
- Capacity-strengthening activities, including webinars, workshops and perspectives on frontier digital innovation.

AoW 4: Enabling Environment
Enabling Environment addresses critical research questions on how organizational innovations, including networks of innovation hubs and centers of excellence can accelerate the development and adoption of inclusive, ethical digital innovations across CGIAR, partners and FLW systems in general. Anticipated outputs include policy innovations, governance frameworks, capacity development and business models necessary to institutionalize FAIR and AI-ready ecosystems while enabling all CGIAR and partners effectively leverage digital solutions.

The innovative nature of this AoW lies in exploring the role of institutional interventions in digitally transforming agriculture R4D organizations. These questions and the results from this experiment will have broader implications for digital transformation of the FLW systems.

Key activities include establishing/strengthening digital core across CGIAR, developing policy frameworks and guidelines for digital innovations, creating partnership and business models, and creating network of digital innovation hubs. The AI hub in Abu Dhabi is the first of these digital innovation nodes which enable CGIAR and partners access world-class AI talent and infrastructure that individual centers cannot access independently. Outputs from Enabling Environment enable faster and multi-functional team collaborations for digital and AI development and deployment, and equitable technology access, contributing to improved agricultural outcomes. Capacity development initiatives and innovation challenges serve as additional acceleration mechanisms.

Domain expertise required includes digital platform architecture and innovation network management. This also requires organizational change facilitation, policy development and governance design, and partnership development across research-for-development contexts. This creates an enabling environment that accelerates digital transformation across CGIAR and partners.

2. Codesign and partnerships

The DTA team has actively engaged with a wide range of partners and stakeholders, particularly the CGIAR Science Programs and Accelerators (SPA), through multiple mechanisms.

Stakeholder engagement
During the Inception Phase, DTA hosted a pre-inception virtual and an in-person inception meetings.

The pre-inception virtual meeting on 12 March 2025 was mainly to introduce DTA design and goals to other SPAs as well as CGIAR staff. Prior to the virtual meeting, a survey was conducted to gather stakeholder expectations. During the pre-inception meeting, breakout sessions were held to deepen understanding of the stakeholder needs, supplementing the survey data.

From 18 to 21 March 2025, DTA held an in-person inception workshop in Abu Dhabi to reflect on the insights from the virtual meeting as well as the survey. DTA also reviewed the ToCs and plan of work and budgets (POWBs) and made key inception-phase decisions. Apart from the DTA core team, participants included representatives from the Genebanks Accelerator, Gender Accelerator, and the Food Frontiers and Innovations science programs.

The DTA team also participated in inception workshops of other SPAs such as Climate Action, Sustainable Farming, Breeding for Tomorrow and Sustainable Animal and Aquatic Food in February and March. These multiple engagements helped DTA develop a well-rounded understanding of both the explicit and emerging demands from stakeholders, especially SPAs. These consultations informed the review of DTA's ToCs both at Accelerator and AoW levels, and the High-Level POWBs submitted in December 2024.

Use case framework and HLOs
As outlined in Section 3 of the DTA's full proposal, DTA is adopting a use case framework to translate SPA demands and insights into co-developed innovations and outputs. By synthesizing insights from the survey, inception meeting, and bilateral discussions, DTA identified nine HLOs which are now reflected in the revised ToC. The DTA team also reviewed ToCs of other SPAs to ensure that DTA's HLOs contribute meaningfully to broader impact pathways of other SPAs.

Broader partnerships and support
Beyond the SPAs and CGIAR centers, DTA has been engaging with:

- Tech companies: Microsoft, Google, Amazon
- Government institutions: Kenya Agricultural and Livestock Research Organization (KALRO), India Meteorological Department (IMD)
- AgTech organizations: Kuza Biashara
- Non-profits: Mercy Corps AgriFin, Digital Green

Additionally, the Government of the Unted Arab Emirates (UAE), a key donor, has pledged $17 million in-kind support to DTA for AI and Machine Learning development.

DTA proposes four high priority use cases, developed in partnership with the UAE and aligned with stakeholder demand:

1. AgriLLM –agriculture-focused large language model
2. Genebanks Use Case –supported by AWS through in-kind contributions
3. Digital Phenotyping –with funding support from Google
4. Water Resources Factory

All four use cases respond to clear demand from both SPAs and external partners like KALRO, IMD, Kuza, Digital Green, and Mercy Corps AgriFin.

3. Theory of Change and MELIA

3.1 Theory of Change

Equitable and ethical digital transformation of FLW systems requires empowering scientists, innovators, Science programs, CG Centers, and other stakeholders with the appropriate capacities, tools, and enabling environments across the digital ecosystem as well as a more intentional engagement of end users and local innovators early in the innovation development journey. When these actors have access to FAIR (Findable, Accessible, Interoperable, and Reusable), AI-ready data, user-friendly platforms, and advanced AI-driven innovative digital tools, they will be able to generate insights more effectively, design innovations, and support decision-making. Complementing this with targeted training and capacity development on responsible and inclusive digital innovation ensures that solutions are not only technically sound but also socially relevant and ethical. The Accelerator will operate through four Areas of Work (AoWs): **Data Ecosystems**, **Action Lab**, **Digital Futures**, and **Enabling Environment**.

Strong digital ecosystem, cross-sectoral partnerships among the stakeholders, and institutional governance frameworks further enable the development and scaling of innovations that sustainably accelerate scientific research. These enabling environment components support the creation of inclusive digital innovation networks and responsible data ecosystems that enable equitable access, knowledge sharing, and collaborative problem-solving. As a result, scientists and innovators are better equipped to develop, deploy, and scale digital solutions that meet the real-world needs of communities and systems they serve. These innovations help to accelerate science, enhance research productivity, and promote equitable digital access across diverse contexts.

The Digital Transformation Accelerator, defines its Theory of Change around four interconnected 2030 Outcomes: (1) **Institutionalization of FAIR Data Principles**, where CGIAR centers and partners fully adopt and apply FAIR (Findable, Accessible, Interoperable, and Reusable) practices; (2) **Functional Partnerships for Digital Innovation**, enabling co-creation through collaborative development and deployment of digital tools; (3) **Widespread Use of DTA Innovations**, reflecting broad uptake of DTA-provided solutions; and (4) **Equitable Access and Adoption**, ensuring that all users—specially smallholder farmers, women, and youth—can access and benefit from digital innovations.

These outcomes are underpinned by Intermediate Outcome (I-OCs), which mark key shifts in capabilities and practices. These include generating **FAIR and AI-ready data**, deploying **open and secure data ecosystems**, enhancing **partner capacity** for responsible digital use, and **adopting governance frameworks** that foster data sharing and innovation. AI-powered tools that boost research, decision-making, and collaboration, along with inclusive, user-centered digital services, directly support innovation use and equity goals. Other I-OCs strengthen awareness of frontier technologies, enable inclusive solution design, promote responsible innovation practices, and encourage policy environments that support digital transformation. Considering the rapidly changing landscape of LLMs and Generative AI, this I-OC will endow CGIAR and partner with the institutional capacity to deal and adapt to these disruptive changes and be better positioned to harness these technologies.

To deliver these I-OCs, DTA provides a suite of High-Level Outputs (HLOs) including co-developed metadata standards, FAIR-compliant data tools, scalable analytics infrastructure, AI-enabled research tools, and training programs. These outputs also include frameworks for ethical AI, innovation hubs, digital business models, and knowledge products, which collectively ensure that innovations are responsibly developed, adopted, and scaled. This structured progression from outputs to intermediate outcomes leads directly to DTA's long-term vision of inclusive, data-driven, and sustainable transformation in agri-food systems.

Derived from the initial proposal, this ToC version incorporates ISDC feedback, a demand-needs analysis of other programs, and insights from reviewing all ToCs across eight science programs and two other accelerators. These adjustments were crucial for ensuring the ToC is both demand-driven and rational.

Theory of Change Statement:

If, scientists, innovators and research partners have access to FAIR, AI-ready data, platforms, and equipped with novel digital tools, ***And***
If, they receive targeted training and capacity support for responsible and inclusive digital innovation, ***And***
If, enabling environments, partnerships, and governance systems are in place,
Then, they will be empowered to deploy ethical, inclusive, and scalable digital innovations that accelerate science and research programs, promote equitable access, and support sustainable development across food, land, and water systems.

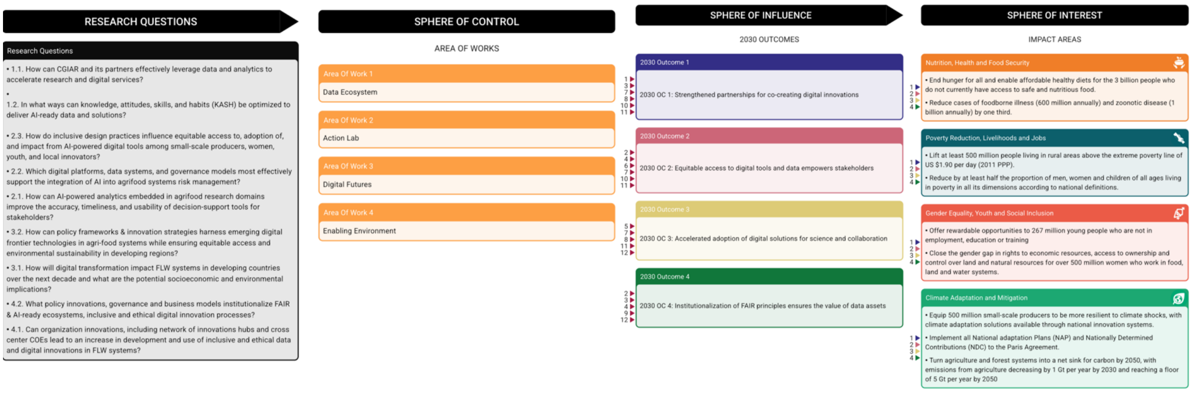
Revision of ToC based on ISDC feedback:

Following feedback from the Independent Science for Development Council (ISDC), the team conducted a comprehensive review of the ToCs for all four AoWs. ISDC's comments highlighted three key areas for improvement:

- Strengthening the linkage between High-Level Outputs (HLOs) and Intermediate Outcomes (I-OCs);
- Making the ToCs more explicit, particularly in identifying and integrating associated risks; and
- Providing greater clarity on delivery pathways and methodologies.

In response, the revised ToCs for all AoWs now address these concerns (find here). The linkages between HLOs and I-OCs have been made more explicit, with clear reflection of how outputs contribute to outcomes and thereby 2030 outcomes. Additionally, each ToC now includes key assumptions and associated risks that underpin the pathways to impact. Specific products and delivery methodologies have also been clearly outlined to enhance the transparency and operational clarity of each pathway. The TOC board is aligned with these changes to communicate with the audience.

It is important to highlight that as the Accelerator begins its implementation, ground-truth insights and learning from delivery processes will inform future iterations of the ToCs, allowing them to evolve and remain responsive to emerging evidence and operational realities.



Accelerator level Theory of Change Diagram

Key Assumptions:

Like other programs, the DTA also relies on several key assumptions—such as the availability of strong partnerships, sustained investment, willingness to adopt innovations, and the presence of enabling ecosystems that support their adoption and use in research. The table below outlines these key assumptions and links them to the relevant Intermediate Outcome (I-OCs).

Key Assumption	Affected HLOs / I-OCs
Assumption 1: Stakeholders are willing and able to adopt FAIR principles and novel digital innovation	I-OC 1.1, I-OC 1.2, I-OC 2.1, I-OC 2.2, I-OC 4.3
Assumption 2: There is sufficient investment and infrastructure for secure, scalable data systems and digital innovation	I-OC 1.2, I-OC 1.4, I-OC 3.1, I-OC 3.2, I-OC 4.1,
Assumption 3: AI tools are aligned with research priorities and regulatory requirements	I-OC 1.3
Assumption 4: Clear governance, data ownership, and consent frameworks are in place	I-OC 3.3
Assumption 5: Ethical guidelines are followed to ensure trust in digital and AI systems	I-OC 3.2, I-OC 4.2
Assumption 6: Decision-makers are capable and willing to use AI-driven insights in policy	I-OC 2.1, I-OC 4.3
Assumption 7: Partnerships are strong and aligned under consistent governance frameworks	I-OC 3.3, I-OC4.3

3.2 *Monitoring, Evaluation, Learning, and Impact Assessment (MELIA)*

The Monitoring, Evaluation, Learning, and Impact Assessment (MELIA) plan for the Digital Transformation Accelerator (DTA) is anchored in CGIAR's Performance and Results Management Framework (PRMF) and follows guidance from the Portfolio Performance Unit (PPU) and the Portfolio Coordination Unit (PCU). It consists of two integrated components: Monitoring, Evaluation, and Learning (MEL), and Impact Assessment (IA). Together, these components are designed to track progress, enable adaptive management, and evaluate DTA's contributions to CGIAR's broader system-level outcomes.

The DTA's Theory of Change (ToC) includes **23 High-Level Outputs (HLOs)**, which feed into **12 Intermediate Outcomes (I-OCs)**, grouped across four Areas of Work (AoWs). These, in turn, contribute to four defined 2030 Outcomes. The relationships between HLOs, I-OCs, and the 2030 Outcomes are systematically mapped in the ToC Board (find here), which outlines key result statements, performance indicators, and delivery targets.

As an accelerator, DTA contributes to CGIAR's impact areas **indirectly—**by enhancing the speed and effectiveness of research, promoting inclusive digital innovation, and supporting the development of a responsible data ecosystem. Therefore, the DTA does not propose standalone impact indicators or independent impact studies. Instead, it contributes to system-level change through collaboration with CGIAR's science programs and other accelerators.

Targets for the HLOs and I-OCs were developed through a consultative process involving discussions between AoW teams and the DTA MELIA focal point. This process ensured that targets are both realistic and aligned with indicator definitions and available evidence on deliverable quality. The causal relationships and targets are provided through the result framework *provided in the attachment*.

Measurement of HLOs is primarily based on **quantitative output indicators**, such as the number of digital tools, platforms, or training events delivered. I-OCs, by contrast, focus on **adoption, integration, and enabling conditions—**including policy uptake, system-level changes, and stakeholder use of innovations. These are also tracked using quantitative indicators but with a stronger emphasis on **influence and uptake**.

The **DTA MELIA focal point** will collaborate closely with **AoW leads** to monitor progress, verify evidence, and report on results, ensuring a coordinated and coherent approach to learning and accountability.

Data Collection Methodology

Given the implementation nature of the DTA, this accelerator will primarily focus on foresight to identify, develop, deploy, and enable an environment around the digital ecosystem. As such, all targets are quantitative. The High-Level Outputs (HLOs) and Intermediate Outcomes (I-OCs) can be directly measured based on the completion and deployment of innovations, as evidenced in completion or annual reports. Furthermore, Key Informant Interviews (KIIs) will be conducted as required to cross-check progress against each HLO and I-OC. These data will be collected on an annual basis for incorporation into the reports.

This approach relies on self-reporting from each Area of Work (AoW), where the AoW is responsible for generating reports that document their progress against the defined HLOs and I-OCs. Supporting documentation and evidence will be preserved to substantiate the reported progress.

Data Quality Control

As noted in the data collection methodology, due to the delivery nature of the accelerator, DTA will only collaborate with other Science projects, Accelerators, CGIAR Centers, and partners. Therefore, surveys are not required. Since these data are counted on a census basis, no estimation is needed. Consequently, the data quality strategy will involve:

- **Evidence Preservation:** Ensuring that relevant documentation (e.g., meeting minutes, published guidelines, tool development reports/logs, training attendance sheets, adoption reports, policy documents, etc.) is systematically collected and stored at the MELIA unit. This evidence will serve as primary verification for reported progress.
- **Reporting Accuracy:** The reports from the AoWs will need to accurately reflect the preserved evidence. This includes an internal review process within each AoW before reports are submitted. KIIs will also be conducted as needed, and their summaries will be recorded.
- **Verification:** This process will involve checking the reported figures against the preserved evidence to ensure accuracy, completeness, and adherence to the defined indicators and targets.

- **Consistency Checks:** Ensuring that the reported numbers align with the definitions of the indicators as provided in the ToC board.

Data Collection Frequency

All data will be collected annually. By the end of every November, all AoW leads will be informed and provided with a template outlining their targets, with a submission deadline at the end of December. The collected data will then be compiled and cross-checked against the information provided. Meetings or Key Informant Interviews (KIIs) will be conducted as required with AoW representatives in early January as part of the quality control process. Following this, the MELIA unit will send the compiled data, including progress and justifications, to the Accelerator Lead for internal approval before incorporation into the PRMS and Type 1 reporting.

4. Comparative advantage analysis

DTA's comparative advantage analysis revealed a strategic landscape where CGIAR's agricultural research expertise and established trust networks complement rather than compete with partners' superior technological capabilities. This analysis fundamentally shaped DTA's partnership-driven approach and resource allocation strategy.

The analysis drove DTA toward a federated partnership model rather than attempting to build all capabilities internally. For moderately positioned HLOs, CGIAR provides agricultural research context, user validation, and stakeholder access while partners contribute technological infrastructure and specialized expertise. This led to flagship initiatives like the AI Innovation Hub in Abu Dhabi, leveraging Abu Dhabi Technology and Research Center (ATRC's) world-class infrastructure that individual CGIAR centers cannot access independently, while CGIAR ensures agricultural relevance and research validation.

The analysis identified three categories of CGIAR positioning across nine HLOs. CGIAR is well-positioned (1.0) in capacity development areas - training for FAIR data ecosystems (HLO4), innovation hub networks (HLO5), and policy development (HLO8) - where agricultural research domain knowledge, established stakeholder relationships, and organizational understanding provide unique advantages. However, six HLOs fall into the moderately positioned category (0.5), including core technical areas like AI-powered products (HLO2), FAIR data innovations (HLO1), and digital infrastructure (HLO6), where partners possess superior technical capabilities but lack agricultural research context and stakeholder trust.

Based on comparative advantage insights, DTA reallocated approximately 60 percent of originally planned infrastructure investments towards partnership development and capacity building. Rather than competing in areas of weak positioning, DTA established strategic partnerships across all moderately positioned HLOs. For example, in AI-powered products development, CGIAR focuses on user-centered design, agricultural ground truth validation, and transparency requirements while partnering with tech companies for technical AI/machine learning capabilities. DTA chose to proceed with all six moderately positioned HLOs because they represent mission-critical capabilities for CGIAR's digital transformation that cannot be outsourced entirely. The analysis demonstrated that while partners excel in technical implementation, they consistently lack three critical elements: deep agricultural research context, established trust with agricultural stakeholders, and commitment to research-for-development outcomes rather than purely commercial goals.

For instance, while tech companies have vastly superior AI capabilities (HLO2), they cannot validate AI outputs against agricultural ground truth or ensure transparency in research contexts, capabilities essential for research credibility. Similarly, cloud providers excel at infrastructure (HLO6) but require CGIAR's guidance on agricultural research-specific requirements and compliance needs.

The partnership strategy transforms potential weaknesses into collaborative advantages, ensuring agricultural research relevance while accessing world-class technical capabilities that neither CGIAR nor partners could deliver independently.

(For further details, please refer to Comparative Advantage analysis attachment)

5. Prioritization

5.1 Knowledge-driven priorities

Building on the Accelerator's rigorous, data-driven prioritization process *as detailed in the Prioritization attachment*, DTA has synthesized evidence across five CGIAR impact areas, the four AoWs, and the nine HLOs to arrive at a finely tuned set of strategic directions. In the first phase, regional and thematic scoring matrices incorporated weighted indicators, from poverty headcounts and data-gap severity to partner readiness and alignment with national priorities, culminating in a ranked list of interventions by region and HLO. In the second phase, these quantitative scores were overlaid with qualitative inputs, partner mandates, funder emphases, existing cross-portfolio synergies, operational capacities, and emerging co-funding opportunities, to produce the final DTA design below.

5.2 Final decisions on DTA's direction and design

The initial scoring crystallized six regional hotspots where targeted AI-in-ag investments promised the highest returns. LAC scored exceptionally high on biodiversity and social-inclusion gaps, driven by underserved Indigenous and Afro-descendant smallholders in Andean and Amazonian zones. As a result, DTA will establish a supporting digital infrastructure in LAC, partner with national research institutes to co-design specialty-crop traceability pilots and embed Indigenous co-researchers as Data Champions.

In CWANA, water-scarcity metrics and private-sector interest in precision irrigation elevated edge-AI irrigation controllers to a top priority for climate adaptation. Although the raw scores for social inclusion were initially modest, donor mandates around gender equity and youth entrepreneurship led DTA to enhance the CWANA portfolio with a gender-responsive irrigation module and recruit local women engineers into prototype co-design teams.

West and Central Africa (WCA) emerged as a critical locus for poverty reduction and social inclusion, with national poverty rates exceeding 40 percent and documented data deserts among women's gardens and youth microplates. Therefore, DTA intends to scale up community-stewarded network, funding five rural districts to deploy multilingual advisory chatbots and localized credit-scoring models. WCA will also host specific activities with Gender Equality and Social Inclusion (GESI) Accelerator to build context-aware decision-support tools.

Eastern and Southern Africa (ESA) scored highest on climate-risk exposure and data infrastructure gaps, prompting the decision to fast-track a federated yield forecast. A phased rollout can begin in Kenya and Tanzania, where existing CGIAR partnerships allow rapid deployment before expanding to Mozambique and Malawi.

South Asia was prioritized for nutrition and food security owing to the large numbers of smallholders in peri-urban and remote regions. Several micro-finance institutions and agricultural extension services in India and Bangladesh have already committed to integrating vernacular disease-diagnosis chatbots and satellite-driven pest alerts into their advisory portals. DTA will provide the data-platform backbone and coordinate localization into major dialects, ensuring broad reach among women-led farmer groups. Finally, Southeast Asia (SEA) displayed acute needs for FAIR-data traceability in aquaculture, livestock and rice systems but also showed strong potential for regional harmonization through the Association of Southeast Asian Nations.

While the numeric prioritization offered clarity, DTA deliberately adjusted allocations in several instances to reflect partner and funder priorities, cross-portfolio synergies, and emerging co-funding prospects. For example, CWANA's irrigation pilots were elevated above their original rank once Gulf-state agritech donors expressed interest in co-investing, and gender-inclusion metrics were weighted more heavily after consultations with major bilateral funders. In South Asia, disease-diagnosis tools were advanced on schedule to align with the Crops-to-Consumer flagship's upcoming trials, despite slightly lower weights in the initial scoring. Conversely, SEA's aquaculture registry was deferred to harmonize with external programs, preventing resource duplication and ensuring interoperability with regional digital-ID frameworks.

Operational capacities in each region also shaped the final roll-out plan. In ESA, limited in-country AI expertise led to a phased approach: first, targeting national research centers in Kenya, then cascading training through a network of data champions in Tanzania and Malawi. In WCA, where telecom infrastructure remains spotty, DTA paired pilot deployments with off-grid energy grants to power edge-AI nodes.

A handful of intentional deviations from the raw rankings underscore DTA's pragmatic flexibility. CWANA's social-inclusion component was substantially increased post-hoc to satisfy donor gender mandates, even though the region's raw bias-audit scores were below average. In SEA, specific secondary biodiversity dashboards were scaled down to single-country pilots, despite high biodiversity-gap scores, because of limited satellite data access and to make room for the delayed aquaculture registry.

Furthermore, recognizing that knowledge-driven scores alone cannot guarantee impact, DTA decided to phase down work in areas lacking local ownership or operational feasibility. For instance, secondary biodiversity dashboards in ESA, which scored moderately on HLO2 but faced data-access constraints, were trimmed back to a pilot in Kenya only, with plans to revisit broader rollouts as remote-sensing capacity strengthens.

By marrying the rigor of quantitative prioritization with the nuance of stakeholder mandates and operational realities and DTA's mandates, the Accelerator ensures that every dollar and person-week invested is channeled to regions, tools, and outcomes where they will do the greatest good. These final decisions, and the underlying scoring matrices and deliberations, are transparently documented in the ***Prioritization attachment.***

6. Alignment of Window 3 and bilaterally funded work

While designated pooled funding will drive much of the Accelerator's early growth, bilateral funding will play a critical complementary role. It will support the scaling of digital and data programs, the development of core infrastructure and governance, and the implementation of specific projects. Bilateral funding will also be essential in sustaining and expanding FAIR and AI-ready data ecosystem that underpins much of the Accelerator's work. As DTA matures, a balanced mix of pooled and bilateral funding will be required to maintain shared digital and data assets and to scale innovations across programs, centers, and partners.

Currently, only 14 bilateral projects across six centers, with a combined budget of $33 million have been mapped to the Digital Transformation Initiative. While some alignment efforts have taken place during the Inception Phase, broader mapping of bilateral projects has been limited for several reasons, including the relatively smaller share and scale of digital-related bilateral investments across CGIAR. In some instances, we identified technically complementary partnerships that align well with the Accelerator's objectives but are not reflected in this mapping, as they do not take the form of bilateral projects involving fund transfers.

As DTA enters its implementation phase, there will be a concerted effort to identify and align relevant bilateral initiatives with the Accelerator's Theory of Change and to establish strategic, synergistic partnerships to maximize impact.

7. Plan of results and budget

(Please refer to DTA's Plan of Result and budget attachment)

The DTA is an accelerator and hence all the centres including World Veg and ICRAF – CIFOR are part of the design, development and delivery of this program. Further, given the central role of digital in CGIAR's strategy, DTA was designed to be quite ambitious in its scope and design. The scope and relevance of DTA is much wider than its predecessor Digital Innovation (DI) initiative, which was the only initiative from the P22 portfolio that was mapped to DTA. This posed a significant challenge since only ten of the fifteen participating centres were part of DI. Large parts of the proposed activities under the Digital Futures (AoW 3) and Data Ecosystem (AoW 1) are novel and include activities and associated outcomes which are not only futuristic but also meant to set up systems and process for CGIAR to take advantage of the AI revolution. Current activities under DI which DTA proposes to continue largely map to Action lab (AoW 2) and Enabling Environment (AoW 4). However, as DTA continues to refine its detailed work plan in 2025 in consultation with other SPAs, DTA will move towards scoping innovative products and tools within AOW 2 and AOW 4 and retiring those activities that don't rank higher as per the prioritization methodology and stakeholder feedback. Considering all the above constraints, DTA's budget tries to balance continuity of critical activities taking cognizance of the capacities of centres to deliver on the DTA mandate while creating space for new activities/AOWs and also ensuring equity across all centers. We ensured that every center receives a minimum allocation of $80,000 primarily to be used for AOW 4 and specifically HLO 4.6. Based on these principles, summary distribution of the budget across various AoWs is shown below:

AoW#	AoW Name	W1/2 allocation	% of Total
AoW 0	Cross-cutting and management	2,296,322	34.59%
AoW 1	Data Ecosystem	534,917	8.06%
AoW 2	Action Lab	1,580,946	23.81%
AoW 3	Digital Futures	714,858	10.77%
AoW 4	Enabling Environment	1,172,957	17.67%
AoW 11	Inception Funds	339,101	5.11%
Total		**$ 6,639,101**	**100.00%**

A reading of the percentage allocation across these AOWs must done in the context of the background explained here. The DTA was allocated W1/2 baseline budget of $4.7 Mn in Dec 2025 which is based on the $4.7 Mn that DI received as W1/2 between 2022 and 2024. Apart from this, $0.339 Mn was earmarked by SO towards inception costs to be incurred between Jan and June 2025. As advised by PCU, the DTA leadership created a high- level plan of work and budget with 79% of the baseline W1/2 budget and earmarked the remaining 21% under the unknown centre bucket. In Feb 2025, DTA's AOW 4 received a designated funding from UAE Govt to the tune of $3.57 Mn for a period of three years. With this designation from UAE Govt, the W1/2 budget for DTA was revised to $6.6 Mn which now included $3.57 Mn earmarked for AOW 4 for three years.

However, the UAE's earmark is mainly to enable DTA to work with UAE Govt to establish and operationalize an CGIAR AI HUB and support the development of AI based scalable and open access digital innovations that support various actors and use cases in the FLW systems in collaboration with UAE based institutions. The outputs and the outcomes that DTA will achieve with these funds is however only a portion of the larger goal and output envisioned for AOW 4. Therefore, to create clear accountability between the UAE earmark and the AOW 4 outputs, a dedicated high-level output (HLO 4.6) to link activities and outputs to be delivered from UAE's earmarked budget has been created and reflected in the PORB. Currently, activities and outputs equivalent to $1.4 Mn dollars from the UAE earmark have been identified and the funding to this extent has been distributed across all the centres. It is pertinent to note that the five centers which are receiving the minimum earmark of $80,000 will be receiving this funding under HLO 4.6. The detailed planning for the remainder of the UAE earmark ($2.08 Mn) is underway, and this amount for now has been reported under AoW 0 cost-cutting and management. AOW 0 also include the cost of the director ($184,200), travel ($30,000), PCU costs and AOW Leadership costs as advised by PCU. The funds under UAE earmark will be reallocated to appropriate AoWs as and when more information is available leading to significant adjustments in the percentage allocation across AOWs. AoW 11 indicates inception funding for the period of January- June 2025. The costs of PCU is distributed across ILRI, CIMMYT and IFPRI while the AOW leadership costs across nearly ten centers.

Given this situation, DTA's budget distribution across AOWs will appear skewed towards AoW2 (Action Lab) and AoW 4 (Enabling environment). However, it's important to recognize that both these AOWs reflect the bulk of continuity of critical work that were part of DI. However, DTA will be intentional about adjusting these allocations, especially on AoW1 and AoW3 based on emerging demands from SPs as well as insights from prioritization. To this end, DTA has active dialogue and conversations with other SPAs as well as centers. To summarize, the DTA's budget process has been an attempt to balance equity across centers, continuity of critical activities from DI while also creating space for new activities, providing seed funding for new AOWs and incorporating outputs that align with donor priorities.

8. Cross-portfolio linkages and geographic coordination

8.1 Cross-portfolio linkages

DTA will work across multiple CGIAR Science Programs and external partners but will link most closely with the Sustainable Farming (SfP), Breeding for Tomorrow (B4T), Sustainable Animal and Aquatic Foods (SAAF), Multifunctional Landscapes (MFL) and Climate Action Science Programs (CASP). These SPs have dedicated AOWs (Eg. AOW 8 in SfP, AOW 6 in SAAF) and corresponding HLOs that demonstrate strong digital and data innovation co-creation opportunities for DTA and hence these five SPs are being referenced in this section. Further, the DTA core team also has been actively contributing as well as leading some AOWs in these SPs. Activities largely focus on standardizing data to render it more AI-ready and rapidly usable. Proposed activities include developing ontologies, digital tools to make data FAIR and AI-ready as well as fostering communities of practice to co-develop governance and policy interventions. This data will be accessible to CGIAR and partner scientists and researchers via DTA's AgPile flagship product to federate data across CGIAR and partner databases. These partner programs will also provide use cases for the DTA's AoW2, focusing on enabling Large Language Models (LLMs) and other cutting-edge approaches to generate decision support solutions that can be scaled in partnership with the Sustainable Farming Science Program. Centers involved in DTA and these Science Programs have begun contributing to the development of human-verified question-answer pairs to be made available for DTA AoW2 AI-mediated approaches such as the AgriLLM (in development through the AI Hub in Abu Dhabi) and for use by partners like Digital Green. Another area of collaboration is to offer use cases to test, validate, and potentially enhance the Multi-Dimensional Inclusivity Index (MDII) to assure digital inclusion, being developed under DTA. All partner programs will also link closely with DTA's AoW3 (Digital Futures) as testing/proving grounds for new technologies of relevance, and AoW4 (Enabling Environment) efforts to enhance CGIAR's data governance and culture.

8.2 Geographic coordination

DTA is – by its nature - not dependent on geographic presence as much as some other fieldwork-based programs might be. It envisages a "Network of Hubs" as part of its Enabling Environment efforts - to assure coherence and communication across programs, partners and the other AoWs. This decentralized network of innovation hubs will efficiently and collaboratively facilitate cross-cutting functions to work with programs and partners in developing demand-driven, user-centric and scalable digital innovations that are geographically validated and appropriate. A hub located at the ATRC will be operational by September 2025. The locations of other hubs are being assessed, but will optimize efficiency, potential for impact, and partner collaboration and demand articulation (e.g., through work with models like the Sustainable Farming Program's Partnerships for Accelerating Agricultural Solutions at Scale (PAASS)). They may be based on specific themes. The Abu Dhabi hub, for instance, is emerging as a global hub for AI. It leverages ATRC's advanced hardware, software, business consulting, and innovation management capabilities, and offers a compelling location to bring together staff and partners across programs.

We anticipate that the Network of Hubs will foster bi-directional communication, collaboration and cross-CGIAR and geographic efficiency, working through existing local networks and models to detect demand, facilitate collaboration, share capacity, and engage with local partners. Thus, the network will operationalize demand and facilitate scaling of promising innovations, ensuring alignment and learning across geographies and with DTA's strategic objectives. It will stimulate innovation within CGIAR and its partner communities through activities including hackathons and a flagship innovation challenge (building on lessons from the Big Data Platform INSPIRE Challenge), and ATRC's expertise in designing and executing such challenges. These activities will encourage the development of digital solutions that are localized, leveraging global technologies and local knowledge.

9. Risk management

Please refer to the following ***link*** with the Accelerator's full risk register. This link is open to ISDC colleagues and others will be granted access, on request.

#	Category	Title	Description	Current Risk Level	Target Risk Level	Actions /Controls to manage risks
1	**Data**	Data integrity	Inconsistent data quality and regulatory environments could affect sustainability of the data ecosystem and governance to support FAIR principles.	20	6	An objective of the Data Ecosystem AoW as well as the Enabling Environment AoW is to build robust institutional and center level systems, capacities and processes to ensure that data that is currently generated is FAIR, AI-ready as well as stored and managed well. We are also creating cross CGIAR COP called data collaborative which will be the vehicle to drive this vision across the centers.
2	**Partners and partnerships**	Stakeholder engagement and adoption	Limited stakeholder buy-in, insufficient incentives or communication gaps could impede adoption and utilization of digital innovations, reducing their impact.	16	6	DTA by design is going to co-develop innovations and outputs in collaboration with other SPs. We are incorporating into our management logic mechanisms that drive the co-development philosophy all through the innovation development lifecycle. Further, DTA is committed to human-centered design (HCD) and GESI in its outputs/outcomes and this reflects in our AOW design as well.
3	**Business continuity**	Data security	Data risks could emanate from potential data leaks, losses as well as violation of sovereign data protection and privacy loss. All these different kinds of risks could lead to legal and reputational damage.	15	4	As part of the accelerator, DTA sub-AoWs will specifically address issues of policies, technologies as well as SOPs that will minimize potential data risks. DTA also has activities that are meant to improve data management and storage and minimize the chances of data loss. Further, most CGIAR centers and projects have sufficient safeguards to ensure that the data risks are minimized.

4	**Data**	Data ownership	Lack of clarity on IP may hinder use of AI-ready data, affecting digital product development.	12	6	As part of DTA deliverables, the Accelerator will also work on developing a policy to specifically address the issue of IP around use of data generated by CGIAR and partners.
5	**Data**	Data culture	Reluctance to require open, FAIR data, to value and incentivize it as a core CGIAR asset could affect efforts towards data-driven impact.	12	6	DTA's AoW4 has several sub-AoWs (especially digital core) that will strive to work with the center leadership teams as well as drive a common vision on data and digital management. One of the key parts of this vision is to also drive change management and advocate for center leadership to value data as an asset and also nudge their investments into transforming their data into a valuable asset.

10. Addressing feedback on select topics

Concise summary of feedback	**Details on how DTA has addressed or will address this feedback** (in the Inception Phase and beyond)
Revise the ToC to better link outputs (e.g., AI) with outcomes (e.g., improved agri-food resilience). Clarify how acceleration, support, and facilitation will be achieved. Make innovation and policy pathways more explicit, as current focus is clearer for capacity development (e.g., training) but lacks detail for innovation and policy	We have significantly revised our TOCs and also constantly made it clear that DTA's pathways to impact will only be through co-creating outputs with other SPAs (Refer to sections 2 and 3). We have also reviewed in detail the ToCs of other SPAs to ensure the HLOs we promise in our ToCs have entry points into the impact pathways of other SPAs. We have revised the language of HLOs and I-OCs to articulate how acceleration, support and facilitation will be achieved.
Be more specific about engagement with other SPAs. Identify joint outputs/use cases, including details on funding, staff time, and geographies. Prioritize use cases in the global South and ensure inclusion of marginalized groups in their selection, development, and deployment.	As articulated in section 2, we have undertaken significant efforts and consultations to achieve more specificity in terms of the joint use cases that could be developed. We also have been working on a more detailed low level work plan which details funding split, and staff time in collaboration with other SPAs. A lot of effort was also made to ensure that the agreed outputs/use-cases are consistent with the revised ToCs and the HLOs and use cases were prioritized based on their relevance and impact for the global south and marginalized groups.
Develop a risk register with clear mitigation measures. Key risks include: (1) lack of CGIAR staff engagement, (2) maladaptive innovation outcomes, (3) high energy use by data centers, (4) data loss from extreme events or cyber-attacks, (5) inadequate safeguards against data monetization, and (6) technological risks tied to AI-readiness and application across the five Impact Areas.	We have revised the risk register and incorporate these in the new risk register mentioned in section 9
Make the MELIA framework more action-oriented	The new MELIA framework which is derived from our ToCs is carrying metrics which are more quantifiable as well as consistent with the proposed outputs and activities.
The following suggestions were made for the four AoWs AoW 1: How will proponents ensure CGIAR staff use and standardize data for the database (e.g., GARDIAN)? AoW 2: Research questions are untestable and need revision. AoW 3: Consider renaming to "Digital Frontier Technologies." AoW 4: Research questions are untestable and need revision.	We have revised the ToCs, objectives as well as PORBs in response to the comments for the four AoWs.